Ultra-high thermoelectric power factor in multi-valley p-type CoVSn half-Heusler from first principles calculations

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Abstract: Half-Heuslers (HH) have been extensively explored as thermoelectric (TE) materials with high power factors (PF). Some of the higher PF materials in general, and half-Heuslers in particular, are materials with high band and valley degeneracies, which provide a large amount of electronic transport pockets. In this work, we computationally study the TE PF of p-type CoVSn HH material, whose valence band structure consists of 2 L-valley bands and a W-band, with a total valley degeneracy of 14. We use Boltzmann transport considering the full energy/momentum/band-dependence of the different states and of the scattering rates. We account for all relevant scattering processes, i.e. acoustic, non-polar optical, and polar optical phonon (POP), as well as ionized impurity scattering (IIS), all described using first-principles extracted parameters. At 300 K we find a maximum p-type PF of 17.6 mW m^{-1} K^{-2} is at carrier concentration of 9.3 $\times 10^{20}$ cm^{-3}. Compared to the average PF of the HHs family of materials, this more than double PF value is a consequence of the large degeneracy providing many transport pockets and of the high dielectric constant of the material, providing screening and reduced scattering. Using supercell calculations, we have also quantified the change in the PF with aliovalent doping and identified Ti as a possible dopant with minimal degradation in the PF.

1 Introduction

Thermoelectric (TE) generators can directly convert heat into electricity via the Seebeck effect, thereby efficiently utilizing waste heat and promoting sustainable energy development.[1,2,3,4,5] The efficiency of the heat to electricity conversion process is quantified by the dimensionless quantity figure-of-merit, zT, which depends on the Seebeck coefficient (S), electrical conductivity (σ), and thermal conductivity ($\kappa = \kappa_l + \kappa_e$) as zT = $S^2\sigma$ T/κ, where T is the absolute temperature, and κ_l and κ_e are the lattice and electronic components of the total thermal conductivity (κ). These quantities are highly adversely interrelated, making it challenging to improve zT.[6] The key to optimizing TE properties is to identify means to enhance the power factor ($S^2\sigma$) and suppressing the κ simultaneously.

Bi_2Te_3-based alloys remain the benchmark thermoelectric (TE) materials for near room temperature operation, but the scarcity of Te makes them expensive and difficult to scale.[7] Half-Heusler (HH) compounds are a rather new entry in TEs, which could offer a way around this. Their flexible crystal structure accommodates many earth-abundant elements, and several compositions perform very well, notably p-type NbFeSb[8,9,10,11] and n-type MNiSn (M = Ti, Zr, Hf).[12,13,14] Among other common TE materials, HHs are known for a high comparative power factors (PF = $S^2\sigma$), but unfortunately also a high lattice thermal conductivity (κ_L), which is a drawback for high efficiency. Some applications, however, such as thermoelectric cooling and on chip micro-harvesting require concentrated heat to flow through the TE material rather nearby materials, thus modules with medium to higher

thermal conductivity are preferred. In these cases a material with a high PF and medium-level thermal conductivity (several W/mK) can extract substantial power from the incident heat flux, even if its figure of merit (zT) is relatively modest.[9] This makes HHs particularly promising candidates for such applications among others.

The main advantage of HHs leading to high PFs, especially for p-type materials, is their large valley degeneracy. For example, the valence band (VB) of NbFeSb, the highest PF performing p-type HH, consists of 2 L-valley bands with a total degeneracy of 8, leading to a PFs around 10 mW m^{-1} K^{-2} at room temperature and gives a peak value of around 11.5 mW m^{-1} K^{-2} at 500K.[10,15] There are HH materials, however, with even higher VB valley degeneracy, involving in addition the Γ-valleys and even more importantly the W-valleys, with an additional degeneracy of 6. CoVSn one such material with two L-valleys and W-valleys with a total degeneracy of 14, making very attractive for high electrical conductivity and PF. It is also an attractive candidate because it is contains comparatively abundant and low-cost elements. It contains no Te, and no Hf or Zr, which are costly and add to supply concerns as is the case for the widely studied MNiSn (M = Ti, Zr, Hf) family. First-principle calculations based on the constant relaxation approximation predict CoVSn to be a stable semiconductor with a power factor comparable to that of the established CoTiSb around 20 mW m^{-1} K^{-2} at 900 K, and a high zT under heavy hole doping, which suggests it can be a candidate for high-temperature power generation,[16,17,18] but this approximation is know to include arbitrary errors.[19] Experimentally, however, the compound is not widely explored since it typically suffers from stability issues and has been realised only recently. For example, Zaferani *et al.* obtained a multiphase sample containing the binary intermetallics SnV_3, Co_2Sn and Co_3V, and these secondary phases degraded the predicted performance and shifted the overall transport from p-type to n-type.[20] On the other hand, it is known that stability can be increased substantially in thin films of materials. For example, Fe_2TiSi is unstable in bulk form, but stable in thin film form[21]. Phase-pure, well-controlled CoVSn therefore remains largely unexplored, but based on its very promising PF, it is worth investigating its ultimate performance either for thin film or bulk TE applications. Ab initio studies on the upper PF level, can also serve as pilot studies for the performance of other multi-valley materials.

In this work, we explore the TE performance of p-type CoVSn HH material and the role of its high band and valley degeneracy. We found that high degeneracy, in combination with the high dielectric constant of this material, leads to ultra-high PF of 17.6 mW m^{-1} K^{-2} at 300 K. This is reached at high carrier concentrations which in practice is reached by aliovalent doping. Such doping optimization could alter the electronic structure, leading to reduction in the maximum achievable PF. Thus, we explore a few possible dopents (Ti, Zr, and Hf), for which the transport calculation is performed on a supercell structure, and show that large valley degeneracy

is more susceptible to degeneracy loss upon doping. We identify, however, Ti as the most appropriate dopant for retaining the high degeneracy of the electronic structure and the high PF.

2 Electronic structure and deformation potential extraction

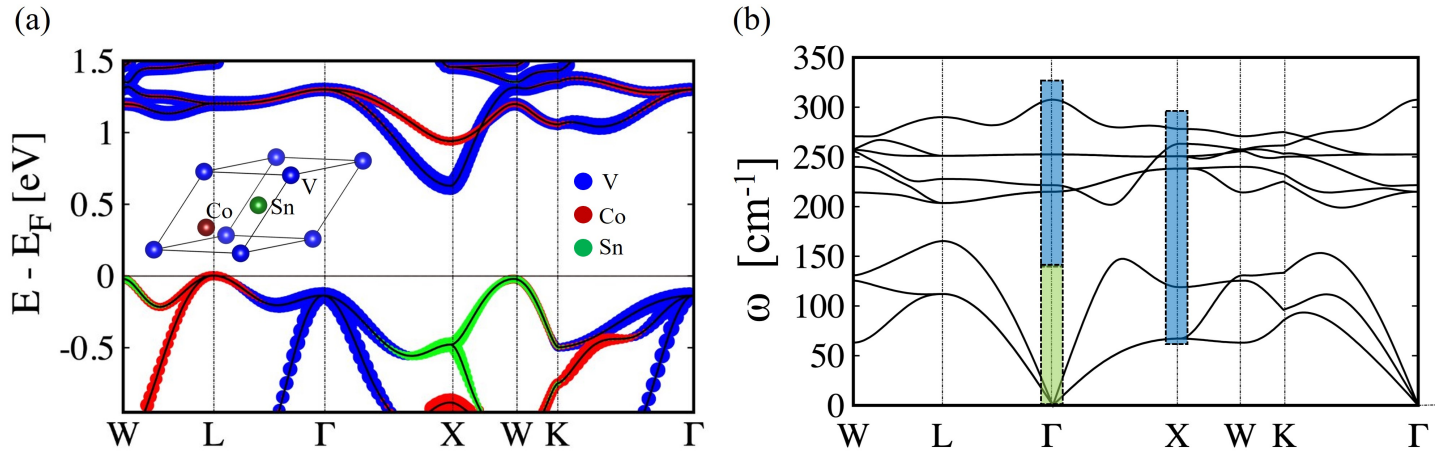


Figure 1: (a) Electronic band structure of VCoSn with atom projected contributions: blue for V, red for Co, and green for Sn. (b) Phonon dispersion of VCoSn. The green shaded region marks approximately the acoustic phonon branch regions used for the electron-phonon coupling matrix element calculation. The the blue shaded region marks approximately the optical phonon branch regions from which the optical phonon matrix elements are computed (for intra- and inter-valley scattering).

CoVSn is one of the rather unexplored HHs because of its critical synthesis condition. It has a small region of stability and impurity phases are more likely to form when synthesized in bulk.[18,22] However, using a modified synthesis condition, it has been reported to form in single phase.[23] CoVSn forms in the HH phase with space group F-43m with Co, V and Sn occupying, the 4c (0.25, 0.25, 0.25), 4a(0, 0, 0), and 4b (0.5, 0.5, 0.5) sites, respectively. The atomic arrangement is shown in the inset of Figure 1a. CoVSn is one of the exceptional HHs to have the valence band (VB) maxima at two high symmetry points, L and W. Figure 1a shows the atomic projected band structure with the VB maxima at the L- and W-points. There are two bands at the L-point each being 4-fold degenerate, and one band at the W-point which is 6-fold degenerate (24 W-points, each residing only by 1/4 in the first Brillouin zone). The total valley degeneracy then becomes N_V = 14. Due to this high degeneracy, the density of states effective mass of CoVSn scales up to m_{DOS} = 7.2 m_e. From the atomic contribution, it is clear that the W-point bands are Sn-contribution determined (green color). The L-point band contribution is Co-determined (red color). Figure 1b shows the phonon dispersion of CoVSn with no imaginary modes, indicating that the material is dynamically stable. The large slope difference (sound velocities) between the LA and TO modes from Γ-L indicates high anharmonicity, which is reflected on the Gruneisen parameter (γ) = 1.97. Having such high anharmonicity will keep the lattice thermal conductivity lower than in typical HHs. The thermal conductivity will be discussed in more detail in Section 4. The highlighted green and blue regions in Figure 1b are rough indications

of the acoustic and optical phonon mode regions that participate in the scattering process, and are considered in extracting the deformation potentials. These are regions that satisfy momentum/energy conservation to facilitate the electronic transitions. The phonons around the Γ-point enable the intra-band/intra-valley transitions, which require only short phonon wavevectors, whereas the regions near the X-point are the ones which facilitate the inter-valley transitions of large wavevectors. The latter are typically weak processes.[10,11,24]

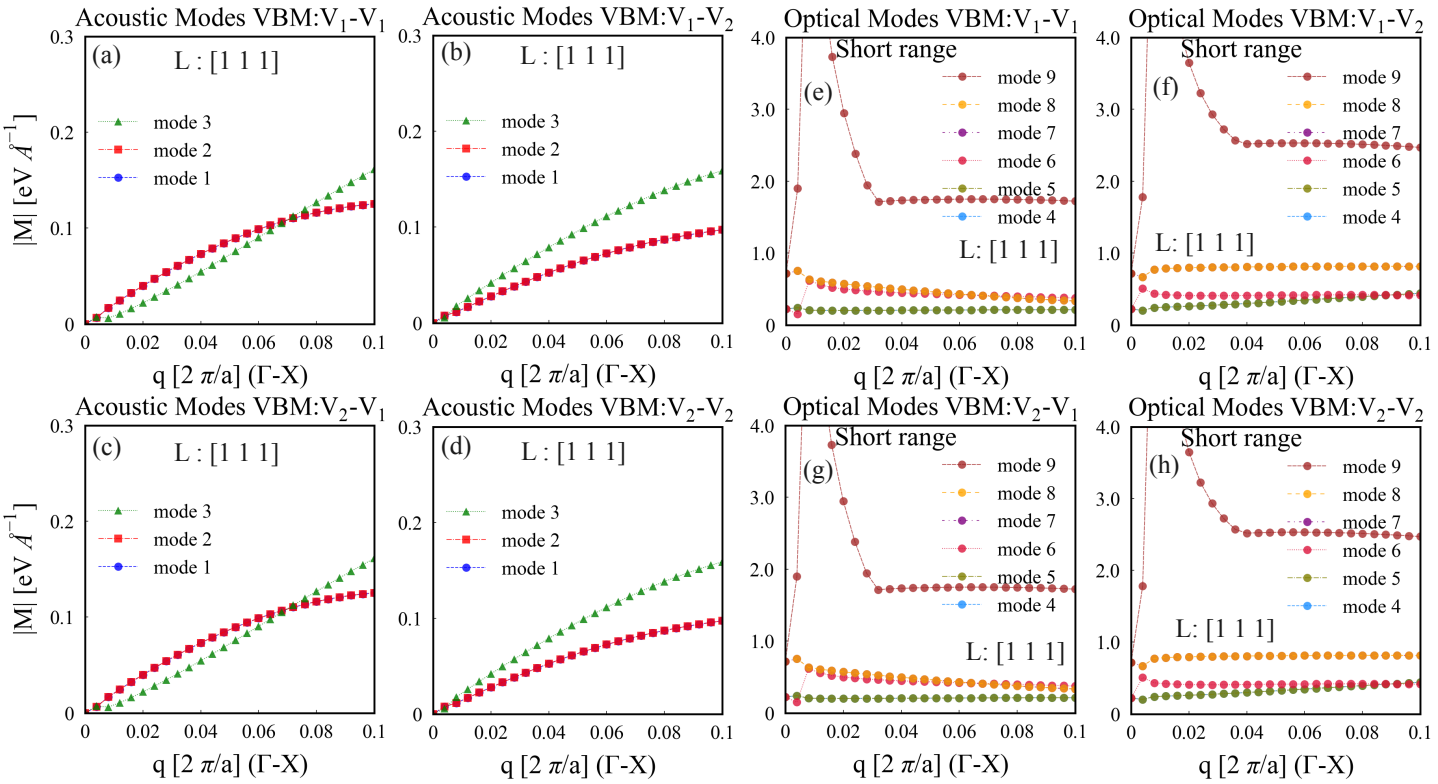


Figure 2: Electron-phonon matrix elements between acoustic phonons and p-type carriers at the two bands of the high-symmetry L-point at [1 1 1], for the following intra-/inter-band transition combinations: (a) band 1 to band 1 (V_1-V_1), (b) band 1 to band 2 (V_1-V_2), (c) band 2 to band 1 (V_2-V_1), and (d) band 2 to band 2 (V_2-V_2). Similarly, the short range matrix elements involving optical phonons from all 6 optimal branches are shown for: (e) V_1-V_1, (f) V_1-V_2, (g) V_2-V_1, and (h) V_2-V_2.

To predict the transport properties accurately for this material, it is essential to carefully consider all different scattering mechanisms that take place. As discussed in the methods, we have included scattering of carriers by acoustic deformation potential (ADP) and optical deformation potential (ODP) phonons, along with scattering from long-range Coulomb interactions: polar optical phonon (POP) scattering, treated using the Fröhlich formalism,[25] and ionized impurity scattering (IIS), treated using the Brooks-Herring formalism. Calculating the first two mechanisms (ADP and ODP) requires the acoustic (D_{ADP}) and optical (D_{ODP}) deformation potentials. In a crude approximation, a global deformation potential can be obtained from the change in band-edge energy with the relative change in crystal volume, $E_{\text{DP}} = \frac{d(E-E_{\text{ref}})}{d(\Delta V/V)}$ [26,27,28,29] (which does not allow separation between acoustic and optical mode scattering, however). Above E_{ref} is the reference energy before strain. Since DFT band structure calculations assume periodic boundary conditions, the absolute position of the energy levels in a

periodic crystal include an unknown constant shift. Therefore, a reference level (E_{ref}) is required to obtain the absolute value of the change in the band edge with respect to volume change. However, an accurate prediction of the deformation potential requires an $ab\ initio$ treatment of electron-phonon coupling.[30,24,10] Here, we have calculated the electron-phonon coupling matrix elements using Wannier interpolation as implemented in the EPW package;[31,32] details are given in the computational methods.

The acoustic deformation potential can be calculated by taking the slope of the matrix elements with the phonon wave vector of the first three branches, $D_{\mathrm{ADP}} = M_{m,n,\nu}(\boldsymbol{k},\boldsymbol{q})/|\boldsymbol{q}|$.[33] Figure 2 (a-d) shows the matrix elements of the acoustic phonons (for carriers at the L high symmetry point, [111]) with respect to the phonon wave vector ($\boldsymbol{q}$) in the Γ-X direction. As the L-point is doubly degenerate (having bands V_1 and V_2), four possible combinations for intra/inter-band/valley scattering emerge, i.e., V_1-V_1, V_1-V_2, V_2-V_1 and V_2-V_2. Matrix elements involving the three acoustic modes are shown, the first two modes being the transverse modes, while the third mode being the longitudinal mode. The overall D_{ADP} is calculated by averaging the transverse and longitudinal deformation potentials as:

$$D_{\mathrm{ADP}} = v_{\mathrm{s}}\sqrt{\left(\frac{D^2_{\mathrm{ADP,LA}}}{v^2_{\mathrm{l}}} + \frac{D^2_{\mathrm{ADP,TA}}}{v^2_{\mathrm{t}}}\right)} \tag{1}$$

where v_{s}, v_{l}, and v_{t}, are the average, longitudinal, and transverse velocities, respectively.

In the case of optical phonons, the deformation potential is the value of the matrix elements themselves $D_{\mathrm{ODP}} = M_{mn\nu}(\mathbf{k},\mathbf{q})$. The D_{ODP} value is calculated by taking the average of the short-range component of all optical modes for small $|\boldsymbol{q}|$-vector values near the Γ point as:

$$D_{\mathrm{ODP}} = \sqrt{\omega_{\mathrm{ODP}}\sum_{\nu}\frac{|M_{mn\nu}(\boldsymbol{k},\boldsymbol{q}) - M^{L}_{mn\nu}(\boldsymbol{k},\boldsymbol{q})|^2}{\omega_{\nu}}} \tag{2}$$

where ω_{ODP} is the average value of optical phonon energy. The short-range optical matrix elements ($|M_{mn\nu}(\boldsymbol{k},\boldsymbol{q}) - M^{L}_{mn\nu}(\boldsymbol{k},\boldsymbol{q})|^2$) are shown in Fig. 2 (e-f). It is to be noted that to extract the optical deformation potential, the long-range component (Coulombic) needs to be subtracted as discussed in previous studies.[30,24,10]

Similar calculations are performed along two other $\boldsymbol{q}-$paths, namely Γ-L and Γ-K, in order to account for the electronic structure and phonon spectrum anisotropy. After extracting the deformation potentials for each direction using the description above, an overall deformation potential value is obtained using the relation:

$$D = \sqrt{\frac{n_{\Gamma-X}D^2_{\Gamma-X} + n_{\Gamma-L}D^2_{\Gamma-L} + n_{\Gamma-K}D^2_{\Gamma-K}}{n_{\Gamma-X} + n_{\Gamma-L} + n_{\Gamma-K}}} \tag{3}$$

Table 1: Deformation potentials for acoustic (D_{ADP}) and optical (D_{ODP}) modes for CoVSn for carriers at the L- and W-points in the VB along different directions and their averages. The 'total' values are the ones used in the simulations.

	Site	Γ-X	Γ-L	Γ-K	total
D_{ADP}	L	1.86	2.31	2.71	
	W	2.6	2.14	2.63	
	overall	2.90	2.38	2.67	2.5
D_{ODP}	L	3.16	1.8	1.72	
	W	1.78	2.04	1.99	
	overall	2.56	1.92	1.86	2.06

Table 2: Input parameters to the BTE for describing scattering in the calculation of the transport coefficients for CoVSn: equilibrium values of the lattice constant; values for acoustic deformation potentials (D_{ADP}); optical deformation potentials (D_{ODP}); the highest energy of the transverse optical phonon mode (TO) at the Γ point, $\hbar\omega_{\mathrm{ODP}}$; the highest energy of the transverse optical phonon mode (LO) at the Γ point, $\hbar\omega_{\mathrm{POP}}$; the high frequency and total dielectric constants, k_∞ and k_0.

Composition	CoVSn
a [Å]	5.8129
density [g/cm^3]	7.729
k_∞	33.37
k_0	51.04
ν_s [m/s]	2987
D_{ADP}(h) [eV]	2.5
D_{ODP}(h) [eV/Å]	2.06
$\hbar\omega_{\mathrm{ODP}}$ [meV]	29.7
$\hbar\omega_{\mathrm{POP}}$ [meV]	37.0

where the square of the deformation potential values are weighted by the degeneracy of each high symmetry Brillouin zone direction.

Here we would like to mention that the methodology adopted for Wannier optimization often leads to different behaviour of Wannier orbitals at the different equivalent high symmetry points, thus, we also perform these calculations for each individual valley and average their deformation potential as:

$$D = \sqrt{\frac{\sum_n D_n^2}{n}} \tag{4}$$

where n is the number of sites corresponding to each high symmetry point in the Brillouin zone, i.e. n = 8 for L-valleys and n = 24 for W-valleys. Detailed calculated matrix elements along different directions and different sites for both L- and W- high symmetry points are given in Supplementary Figures S(10-23).[34] The calculated deformation potential values for ADP and ODP for different directions and the total values used in the simulations are given in Table 2. They turn out to be very similar to the values obtained with many other HH materials.[11]

3 Electronic transport and thermoelectric power factor

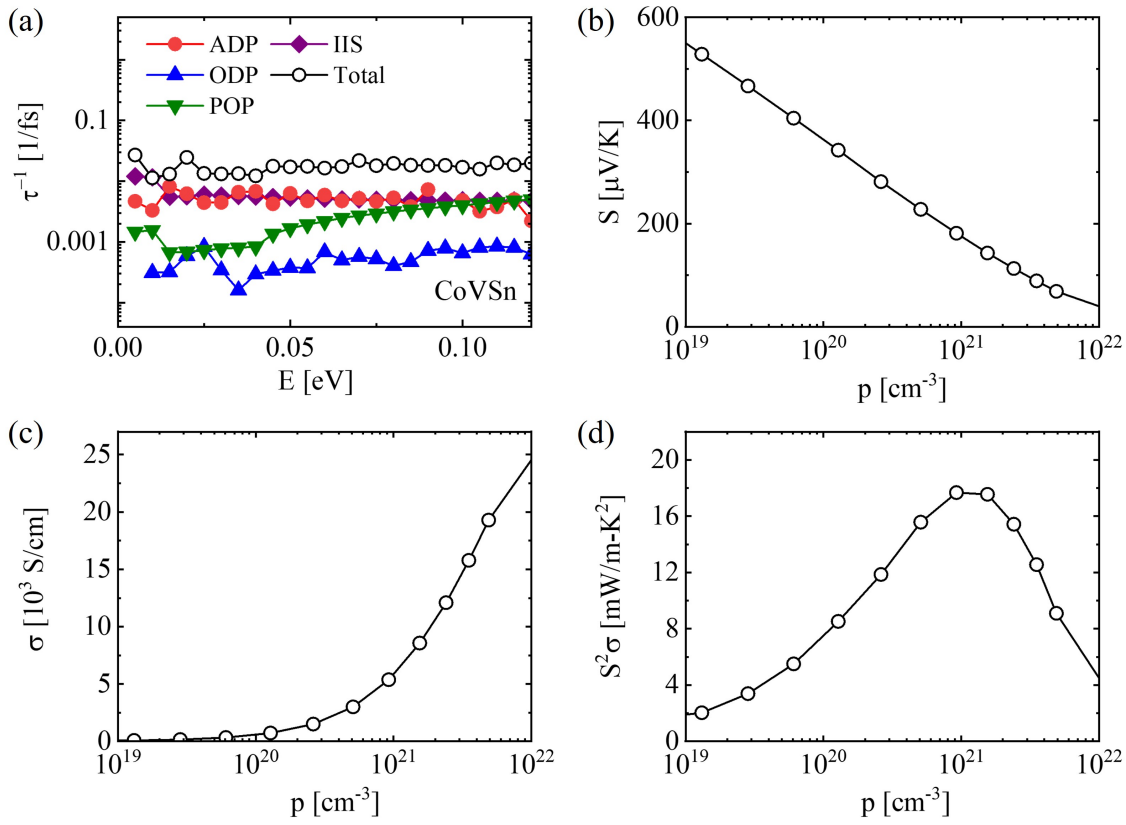


Figure 3: (a) Scattering rates versus energy for different scattering mechanisms in CoVSn at $E_F = 0$ (b) Seebeck Coefficient, (c) electrical conductivity, and (d) PF versus carrier concentration at 300.

We use the Boltzmann transport software ElecTra to compute the electronic transport and thermoelectric coefficients of CoVSn.[35] We consider all important scattering mechanism, i.e. ADP, ODP, POP and IIS. A comparison of the scattering strength between the different scattering mechanisms versus energy for the case where the Fermi level is placed at the valence band edge, is shown in Figure 3a. IIS and ADP scattering are the dominant scattering mechanisms in this material. IIS is typically strong in HHs and other TEs due to its Coulombic nature and the high doping densities,[10,11] but typically ADP is weaker because it is elastic and its scattering phase space is limited. In this case, however, ADP is strong because of the very close proximity of the W-points and L-points, which are only 0.25 % of the BZ apart, which fall between the phase space reach of the ADP process, thus increasing ADP scattering. On the other hand, IIS and POP are weaker compared to typical HHs because of the higher dielectric constant of this material and increased screening due to the higher carrier density as a result of the larger degeneracy.

The thermoelectric coefficients are shown in Figure 3b-d. Figure 3b shows the Seebeck coefficient. The high degeneracy (N_V = 14), results in a high density of states (DOS) and finally a high Seebeck coefficient of about 180 μV/K at a carrier concentration of 10^{21} cm^{-3} (where the PF peaks). Figure 3c shows the electrical conductivity

as a function of carrier concentration. At high carrier concentrations the electrical conductivity is also high, again because of the high degeneracy introducing many transport channels. Note that a degree of inter-valleys scattering between these channels exists, while limits conductivity, but not enough to negate the effect of high degeneracy on transport. With both, high Seebeck coefficient and high conductivity, a maximum PF of 17.6 mW m^{-1} K^{-2} is reached at a carrier concentration of p = 9.31 $\times$ 10^{20} cm^{-3}. Unlike other HH materials, where the maximum PF is achieved around carrier concentrations of the order of 1-4x10^{20} cm^{-3}, in CoVSn this is reached at an order of magnitude higher due to its high degeneracy, and higher DOS, which shifts the power factor peak at higher concentrations.[10,11]

At that carrier concentration, and at peak PF, the Seebeck value is 181 μV/K, and the conductivity is 5364 S/cm. In comparison, the average Seebeck and conductivity in HHs is 156 μV/K and 3140 S/cm.[11]Thus, it is the high Seebeck coefficient as well as conductivity which leads to such high PF. Note, however, that in order to achieve such high carrier concentration, a higher doping concentration required. Such large doping can change the electronic structure and, therefore, the maximum achievable PF. In the next section we explore a few aliovalent doping substitutions and their effect on the electronic structure and PF of CoVSn.

4 Effect of aliovalent doping on electronic transport and power factor

As mentioned in a pervious section, the VB maxima has contribution from Co at the L-point and Sn at the W-point. Therefore, when intuitively choosing a dopant, doping at the V site could have the least influence on the band structure (although in reality indirect contributions would also bring changes). Nevertheless, as a first doping approach, we dope at the V site. To incorporate doping, we use a 2$\times$ 2 $\times$ 2 (24-atom) supercell, dope that, and use its modified band structure in transport calculations. The electronic transport method to calculate the PF using supercells is explained in detail in REf.[36] Before applying any doping, as a validation to our method, the PF calculated using the primitive cell or the supercell for undoped structures is verified to very similar as shown in Figure 4a (black lines, empty versus full circles). We then consider Ti, Zr, and Hf as suitable dopants at V site. Each of the dopants bring one hole into the matrix. In the 24-atom supercell, one atom out of 8 V-atoms is replaced by Ti/Zr/Hf to obtain a doping concentration of 12.5%. By computing the defect formation energy (inset of Figure 4a), we find that Ti is more energetically favorable (lower height red bar). The reason is because of the similar size of V and Ti atoms. With Ti/Zr/Hf doping, however, there is loss in the CoVSn degeneracy. Figure 4b shows the band structure of the undoped CoVSn supercell, in which N_v = 14. After Ti, Zr and Hf doping, the degeneracy has decreased to 6, 3, and 3, respectively, as shown in Figures 4b,c,d. Note that we count

hte degeneracy by the number of bands that are reatained in an energy windiw of 50 meV ($2k_BT$) as shown by the green bars. The reduction in degeneracy brings PF reductions as well, as shown in Figure 4a (red, blue, green lines and symbols). The loss in degeneracy is lowest for Ti doping which shows the highest PF among all. The colored lines in Figure 4a are computed using the band structures to the right. However for each of the colored lines, only one Fermi level position is valid, thus only one point in those figures is valid, the one which corresponds to the position of the Fermi level upon 12.5 % doping. Here the Fermi energy is pushed at 0.1 ev deep into the VB, which will introduce a carrier concentration of 3.5 $\times$ 10^{21} cm^{-3}, much higher than the optimum (9.31 $\times$ 10^{20} cm^{-3}). These are shown by the highlighted colored symbols, indicating a non optimal Fermi level position and reduced PF. Importantly, however, the Ti-doped structure (red point) still resides on the pristine PF line, whereas the Hf- and Zr-doped structures show lower PFs compared to the pristine PF line. It is possible to reach a higher PF with one third of the Ti doping, which will also reduce the degeneracy loss.

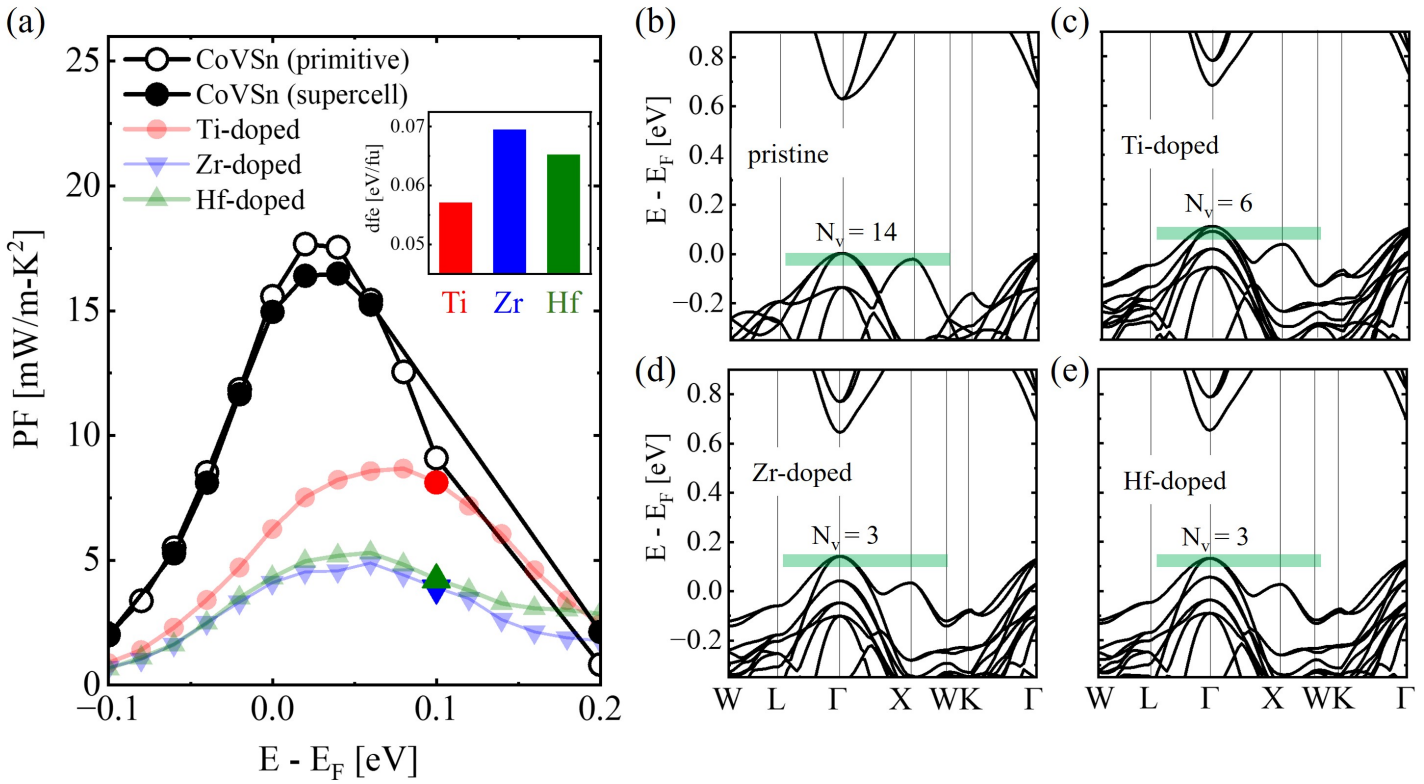


Figure 4: (a) PF as a function of Fermi energy for pristine CoVSn (black open circles) and its supercells (pristine supercell, black with filled circles), and supercells doped with Ti (red), Zr (blue), and Hf (green) at 300 K. The defect formation energy (dfe) of Ti/Zr/Hf doping is shown in the inset. The highlighted colored points are the relevant ones for 12.5 % doping, i.e. one atomic substitution in our supercell. Band structures of supercells: (b) pristine CoVSn, (c) $CoV_{0.875}Ti_{0.125}Sn$, (d) $CoV_{0.875}Zr_{0.125}Sn$, and (e) $CoV_{0.875}Hf_{0.125}Sn$. The green shaded region in (b-e) is the energy window of 50 meV ($2k_BT$). The Fainted data in (a) show the PF as a function of Fermi energy considering these band structures. However, the actual Fermi energies for these doped systems are at 0.1 eV, as shown by the highlighted data points.

5 Lattice thermal conductivity and zT

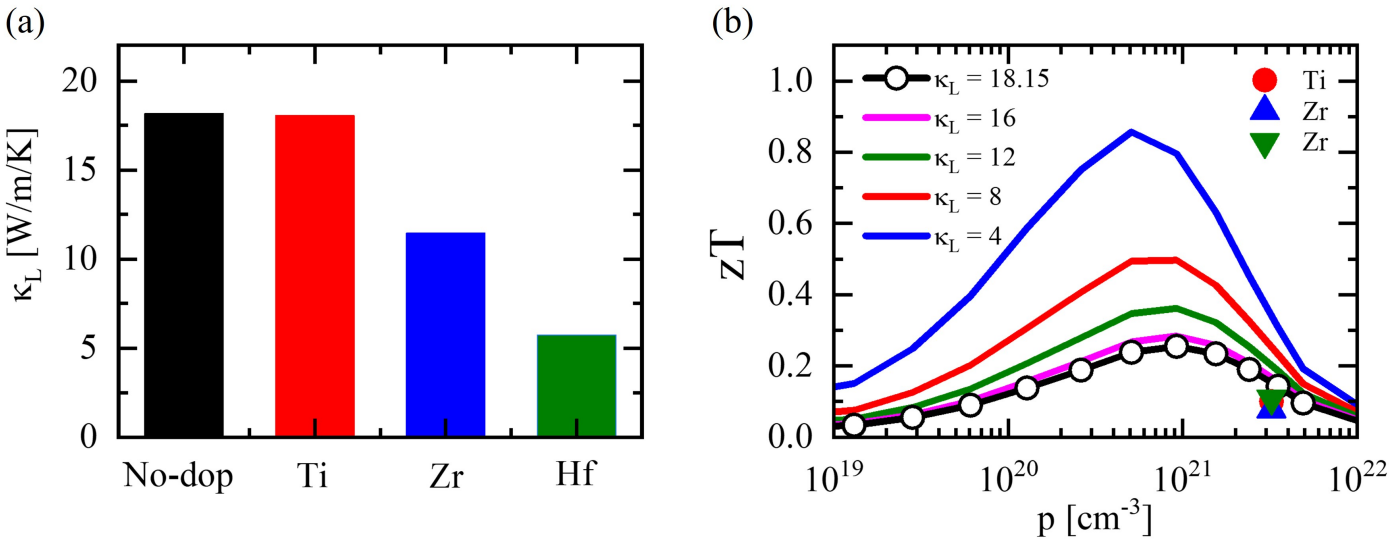


Figure 5: (a) Lattice thermal conductivity of pristine CoVSn (black), $CoV_{0.875}Ti_{0.125}Sn$ (red), $CoV_{0.875}Zr_{0.125}Sn$ (blue), and $CoV_{0.875}Hf_{0.125}Sn$ (green). (b) zT as a function of carrier concentration for pristine CoVSn considering various κ_L along with the data points corresponding to $CoV_{0.875}Ti_{0.125}Sn$, $CoV_{0.875}Zr_{0.125}Sn$, and $CoV_{0.875}Hf_{0.125}Sn$.

However, having similar size and mass to V, Ti doping will not promote strong phonon scattering. Zr and Hf, on the other hand, having a larger mass, can scatter phonons more effectively are reduce thermal conductivity more, improving zT. Thus, despite having a lower PF, the zT can still be significant in case of Zr and Hf doping. To investigate this, we calculated the lattice thermal conductivity and zT of these cases.

Figure 5a shows the lattice thermal conductivity (κ_L) for the pristine CoVSn (no-dop) at room temperature, which is calculated to be 18.15 W m^{-1} K^{-1} . Further to this, in the doped systems we use the Klemens model to incorporate phonon scattering due to the resultant mass disorder.[37,38] In the case of Ti-doping, as Ti has similar mass to V, doping does not create effective scattering centers and the thermal conductivity for $CoV_{0.875}Ti_{0.125}Sn$ remains at κ_L = 18.04 W m^{-1} K^{-1} , almost identical to the pristine value. However, for Zr- and Hf-doping, κ_L is reduced to 11.48 and 5.73 W m^{-1} K^{-1} , respectively. Figure 5b shows the zT of pristine CoVSn (black line, empty circles) with κ_L = 18.15 W m^{-1} K^{-1} , and compared to various hypothetical κ_L cases down to 4 W m^{-1} K^{-1} . The pristine material has maximum zT of around 0.2, whereas to reach zT close to unity, κ_L lower than 4 W m^{-1} K^{-1} is needed, while still maintaining the same PF.

The zTs for all the doped systems are almost of similar values as shown by data points Figure 5b. The reduction of the PF in the Zr and Hf cases is compensated by the reduction in κ_L, resulting in the same zT as that of the Ti-doped structure. Thus, this reveals a trade off between the PF and κ_L in cases of Ti and Zr/Hf leading, to the same zT. A more detailed calculation is required with smaller doping concentration to confirm the best doping

in CoVSn, however that would require a much larger supercell, which makes the calculation more demanding. From the PF point of view, Ti doping of 3-4 % will be the most optimal to reach the highest PF with the tradeoff of retaining the high κ_L of the pristine material and achieve zT close to 0.2.

6 Summary and Conclusions

This work investigates the thermoelectric performance of p-type CoVSn half-Heusler, which exhibits a high valence-band degeneracy of 14. Using first-principles-derived scattering parameters and full energy-, momentum-, and band-dependent Boltzmann transport calculations, we predict a maximum power factor of 17.6 mW m^{-1} K^{-2} at 300 K. This more than doubles the average power factor of half-Heusler materials, owing to its high valley degeneracy and dielectric constant, which enhance transport and reduce scattering. Supercell calculations further identify Ti as a promising aliovalent dopant that minimally degrades the power factor.

7 Computational methods

First-principles calculations within the framework of density functional theory (DFT) were carried out using the Quantum ESPRESSO package. The interaction between valence electrons and ionic cores was described using ultrasoft pseudopotentials. The valence electronic configurations considered were $3s^2 3p^6 3d^3 4s^2$ for V, $3d^8 4s^1$ for Co, and $5s^2 5p^2$ for Sn. A plane-wave basis set was employed with kinetic energy cutoffs of 65 Ry for the wave functions and 650 Ry for the charge density. Exchange–correlation effects were treated using the generalized gradient approximation in the Perdew–Burke–Ernzerhof (GGA-PBE) form.

The defect formation energy (dfe) is calculated using following equations:

$$dfe = E_{CoV_{1-x}(Ti/Zr/Hf)_xSn} - (1-x)E_{CoVSn} - xE_{(Ti/Zr/Hf)CoSn} \tag{5}$$

where E is the total energy.

We use our BTE simulator ElecTra for calculating the transport properties pristine and supercell structures.[35,36] The scattering rate expressions for different scattering processes used are as follows. For acoustic deformation potential (ADP):

$$|S^{\mathrm{ADP}}_{\mathbf{k},\mathbf{k}'}| = \frac{\pi}{\hbar} D^2_{\mathrm{ADP}} \frac{k_{\mathrm{B}} T}{\rho v_s^2} g_{\mathbf{k}'}(E) \tag{6}$$

where k and k' are the initial and final state of carriers, ρ is the mass density, v_s is the average sound velocity, and g(E) is the density of states of the final scattering state.

For optical deformation potential (ODP):

$$|S^{\text{ODP}}_{\mathbf{k},\mathbf{k}'}| = \frac{\pi D^2_{\text{ODP}}}{2\rho\omega}\left(N_\omega + \frac{1}{2} \mp \frac{1}{2}\right) g_{\mathbf{k}'}(E \pm \hbar\omega) \tag{7}$$

where ω is the dominant frequency of the optical phonons, considered to be constant over the entire reciprocal unit cell. N_ω is the phonon Bose-Einstein statistical distribution and the '+' and '-' signs indicate the emission and absorption processes.

For inter-valley scattering (IVS), a similar expression as ODP is used:

$$|S^{\text{IVS}}_{\mathbf{k},\mathbf{k}'}| = \frac{\pi D^2_{\text{IVS}}}{2\rho\omega}\left(N_\omega + \frac{1}{2} \mp \frac{1}{2}\right) g_{\mathbf{k}'}(E \pm \hbar\omega) \tag{8}$$

where ω is the phonon frequency associated with the corresponding inter-valley scattering processes.

For polar optical phonon (POP) scattering, we use the Fröhlich formalism as:

$$|S^{\text{POP}}_{\mathbf{k},\mathbf{k}'}| = \frac{\pi e^2\omega}{\epsilon_0}\frac{|\mathbf{k}-\mathbf{k}'|^2}{(|\mathbf{k}-\mathbf{k}'|^2 + \frac{1}{L_{\text{D}}^2})^2}\left(\frac{1}{k_\infty} - \frac{1}{k_0}\right)\left(N_\omega + \frac{1}{2} \mp \frac{1}{2}\right) g_{\mathbf{k}'}(E \pm \hbar\omega)\left\langle I^2_{\mathbf{k},\mathbf{k}'}\right\rangle \tag{9}$$

where e is the electronic charge, and ω is the dominant frequency of polar optical phonons over the whole Brillouin zone, which has been validated to be a satisfactory approximation.[24] k_0 is the static dielectric constant and k_∞ is the high-frequency dielectric constant and L_{D} is the screening length for POP given by $L_{\text{D}} = \sqrt{\frac{k_\infty\epsilon_0}{e}\left(\frac{\partial n}{\partial E_F}\right)^{-1}}$. Here, $I_{\mathbf{k},\mathbf{k}'}$ is the overlap integral, which is given by:

$$I_{mn}(\mathbf{k}, \mathbf{k}+\mathbf{q}) \;=\; \left\langle u_{m,\mathbf{k}+\mathbf{q}} \mid u_{n,\mathbf{k}}\right\rangle, \tag{10}$$

and is computed directly from the DFT wave functions stored in the Quantum ESPRESSO output.

For ionized impurity scattering (IIS), we use the Brooks-Herring model as:

$$|S^{\text{IIS}}_{\mathbf{k},\mathbf{k}'}| = \frac{2\pi}{\hbar}\frac{Z^2e^4}{k_0^2\epsilon_0^2}\frac{N_{\text{imp}}}{(|\mathbf{k}-\mathbf{k}'|^2 + \frac{1}{L_{\text{D}}^2})^2} g_{\mathbf{k}'}(E)\left\langle I^2_{\mathbf{k},\mathbf{k}'}\right\rangle \tag{11}$$

where Z is the electric charge of the ionized impurity (we assume Z = 1 here), N_{imp} is the density of the ionized impurities, and L_{D} is the generalized screening length given by $L_{\text{D}} = \sqrt{\frac{k_0\epsilon_0}{e}\left(\frac{\partial n}{\partial E_F}\right)^{-1}}$, where E_{F} is the Fermi-level and *n* is the carrier density.

The overall transition rate,$|S_{\mathbf{k},\mathbf{k}'}|$, is calculated by combining the strength of all scattering mechanisms using Matthiessen's rule as:

$$|S_{\mathbf{k},\mathbf{k}'}| = |S^{\text{ADP}}_{\mathbf{k},\mathbf{k}'}| + |S^{\text{ODP}}_{\mathbf{k},\mathbf{k}'}| + |S^{\text{IVS}}_{\mathbf{k},\mathbf{k}'}| + |S^{\text{POP}}_{\mathbf{k},\mathbf{k}'}| + |S^{\text{IIS}}_{\mathbf{k},\mathbf{k}'}| \tag{12}$$

After calculating the transition rates, the thermoelectric coefficients, namely the electrical conductivity, σ, and Seebeck coefficient, S, are calculated as:

$$\sigma_{ij} = e^2 \int_E \Xi_{ij}(E)\Big(-\frac{\partial f_0}{\partial E}\Big)dE \tag{13}$$

$$S_{ij} = \frac{ek_{\mathrm{B}}}{\sigma_{ij}} \int_E \Xi_{ij}(E)\Big(-\frac{\partial f_0}{\partial E}\Big)\frac{E - E_{\mathrm{F}}}{k_{\mathrm{B}}T}dE \tag{14}$$

Here, f_0 is the equilibrium Fermi-Dirac distribution function, $\Xi_{ij}(E)$ is transport distribution function (TDF) defined as:

$$\Xi_{ij}(E) = \int_E \tau_{\mathbf{k},\mathbf{k}'}(E)v_{ij}^2(E)g(E) \tag{15}$$

where, $v(E)$ is the band structure velocity, $g(E)$ is the density of states at energy E, and i, j are the Cartesian coordinate indexes, (for which we set $i = j = x$) and $\tau_{\mathbf{k},\mathbf{k}'}$ is the scattering relaxation time (inversely proportional to the transition rates), which is calculated for each scattering mechanism (m'_{s}) as:

$$\tau_{\mathbf{k},\mathbf{k}'} = \frac{1}{(2\pi)^3}\sum_{\mathbf{k}'} |S^{m'_s}_{\mathbf{k},\mathbf{k}'}|\left(1 - \frac{v_i(\mathbf{k}', n')}{v_i(\mathbf{k}, n)}\right). \tag{16}$$

where the velocities that appear in the bracket are the directional velocities along the assumed transport direction.

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Supplementary Information:

Ultra-high thermoelectric power factor in multi-valley p-type CoVSn half-Heusler from first principles calculations

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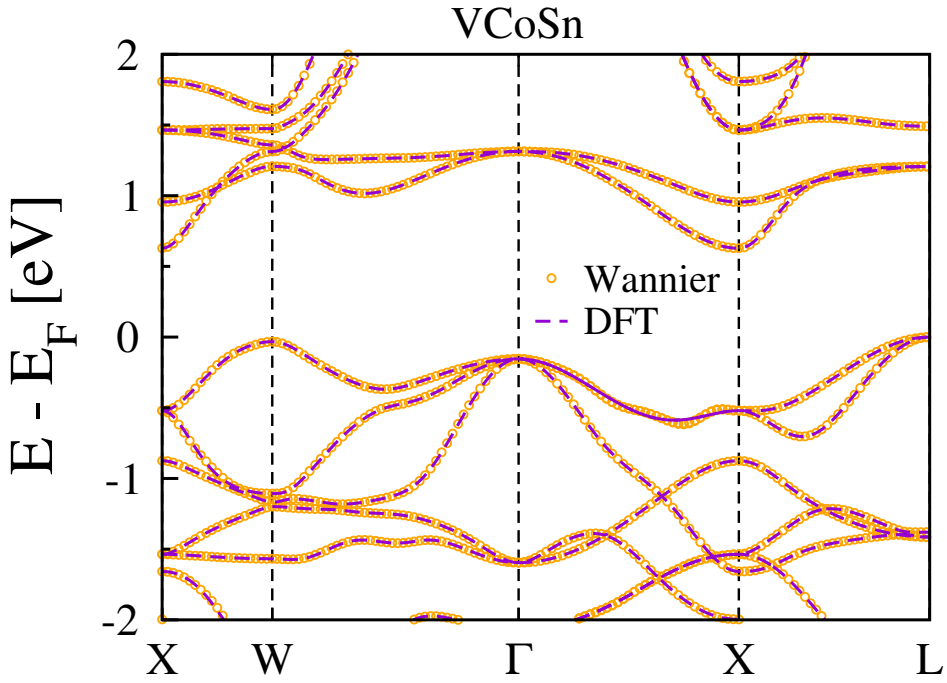


Figure 6: Comparison between the bandstructure obtained from standard DFT calculations and via Wannierization for VCoSn.

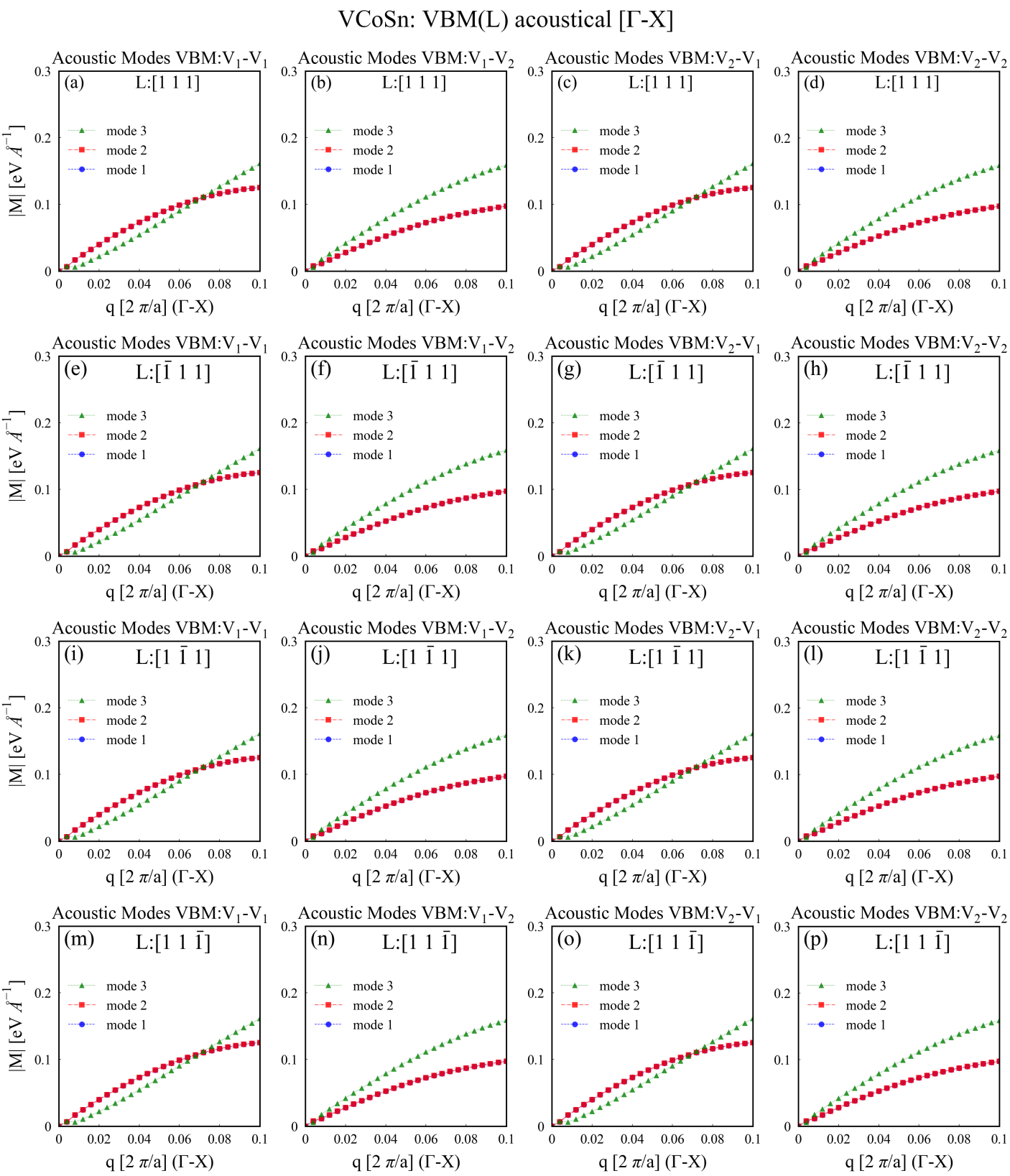


Figure 7: Matrix elements for the VB of CoVSn between the three acoustic phonon modes and L-valley carriers. The phonon wavevectors are in the Γ-X direction, corresponding to intra- and inter-band transitions (each of the degenerate L-points has two valleys from two different bands). All possible interactions between bands 1 and band 2 are presented: i.e. interactions of V_1 with V_1 and V_2. Four high-symmetry L-points are considered: (a-d) [1 1 1], (e-h) [$\bar{1}$ 1 1], (i-l) [1 $\bar{1}$ 1], (m-p) [1 1 $\bar{1}$].

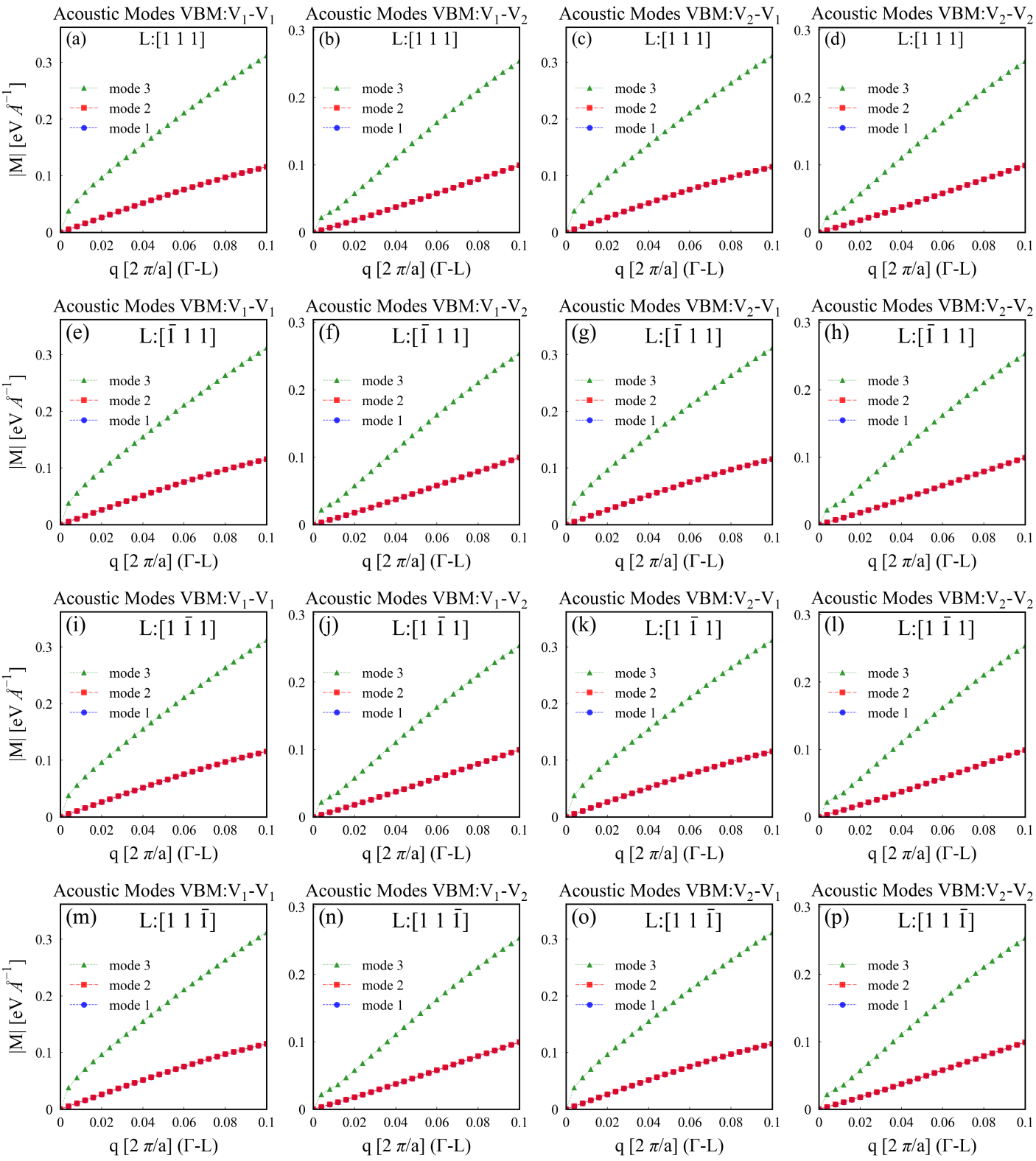


Figure 8: Matrix elements due to interactions of carriers with acoustic phonons for VCoSn around the VBM (at L) in the Γ-L direction, corresponding to intra- and inter-band transitions. The interactions of all band combinations of V_1 with V_1 and V_2 are shown. The four different positions of the high-symmetry L-point are considered: (a-d) [1 1 1], (e-h) [$\bar{1}$ 1 1], (i-l) [1 $\bar{1}$ 1], (m-p) [1 1 $\bar{1}$].

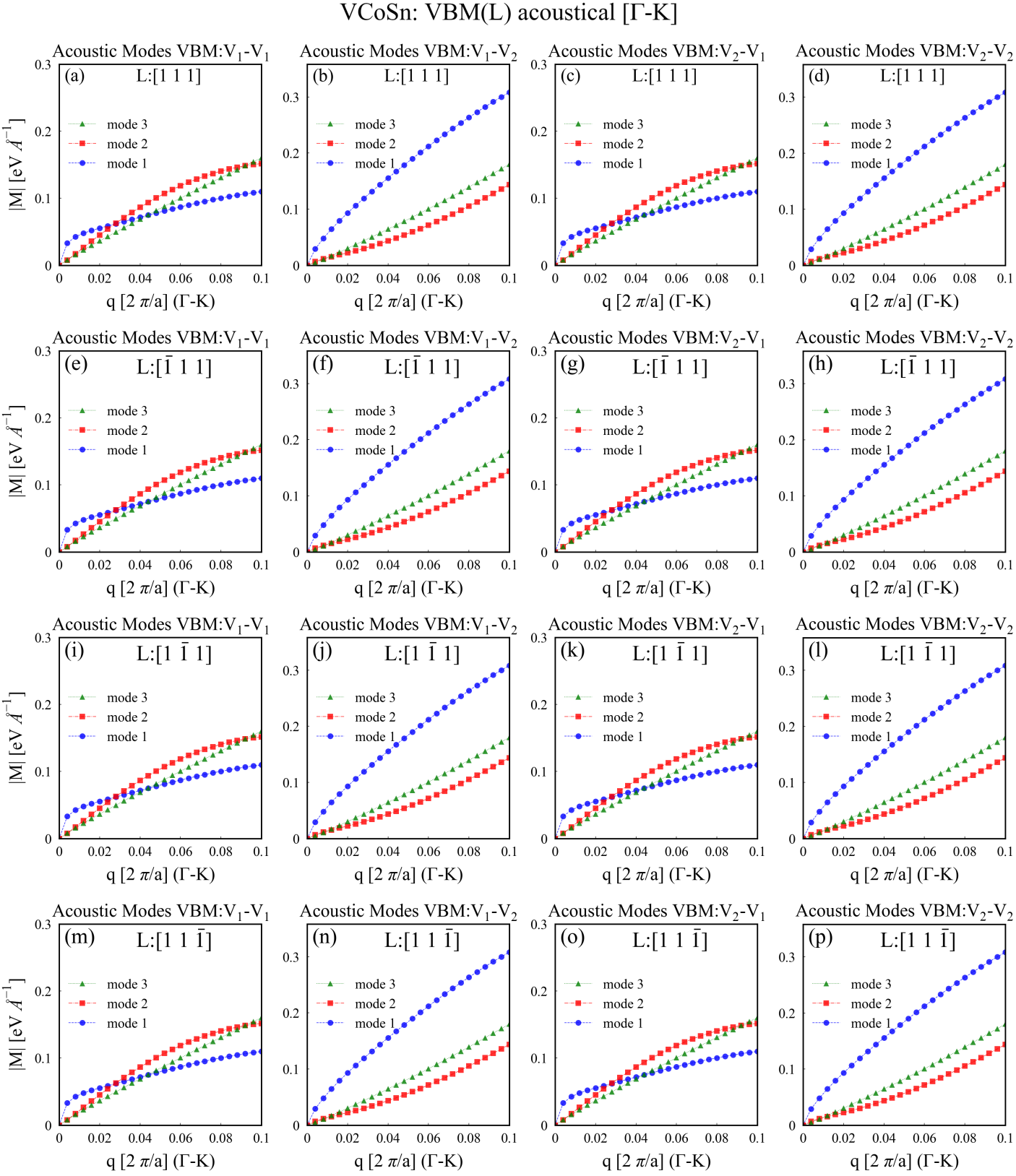


Figure 9: Matrix elements due to interactions of carriers with acoustic phonons for VCoSn around the VBM (at L) in the Γ-K direction, corresponding to intra- and inter-band transitions. The interactions of all combinations of the two bands at L are shown, i.e. bands V_1 with V_1 and V_2. The four high-symmetry L-points are considered separately, i.e.: (a-d) [1 1 1], (e-h) [$\bar{1}$ 1 1], (i-l) [1 $\bar{1}$ 1], (m-p) [1 1 $\bar{1}$].

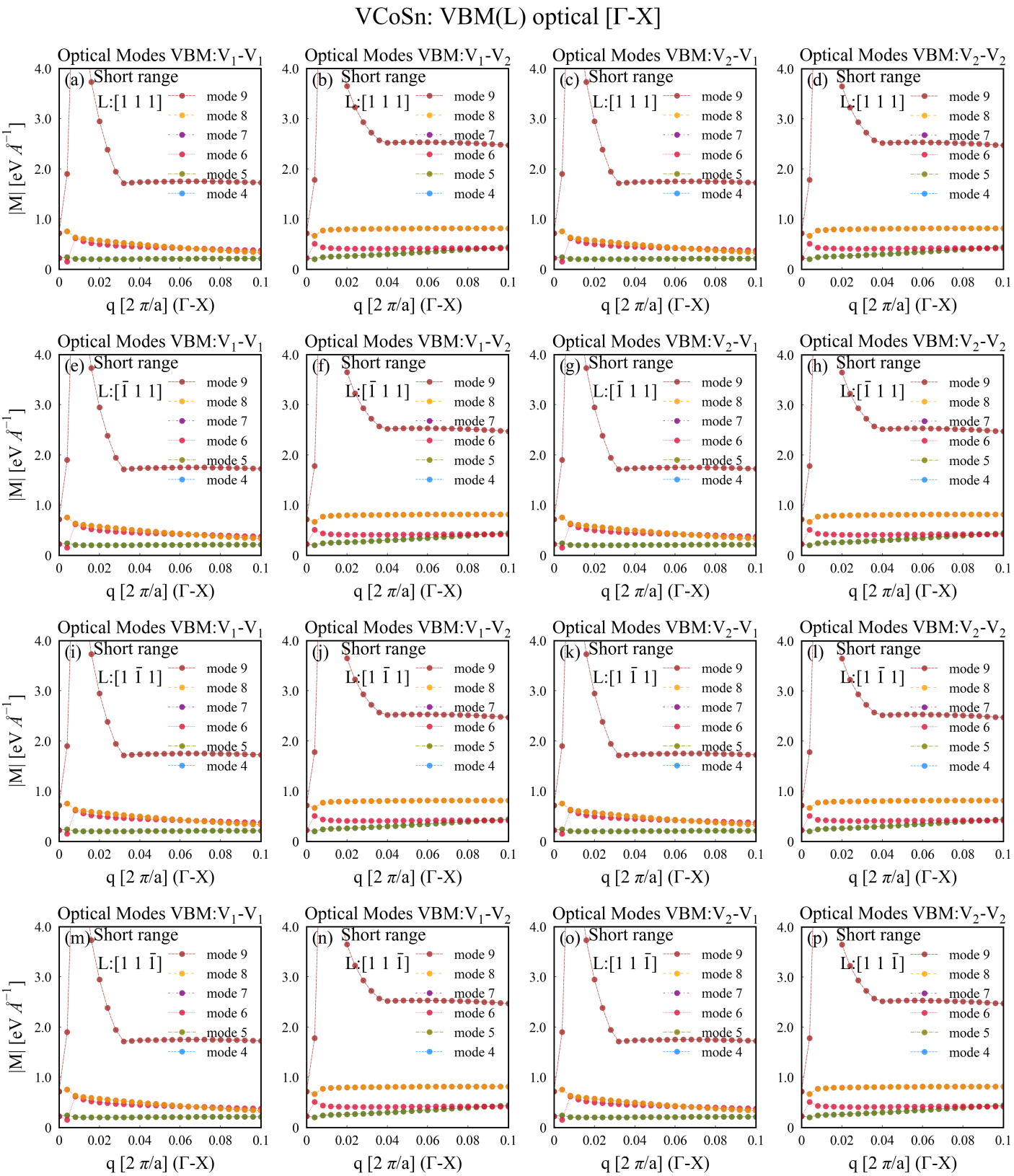


Figure 10: Short range matrix elements due to interactions of carriers with optical phonons for VCoSn around the VBM (at L) in the Γ-X phonon mode direction, corresponding to intra- and inter-band transitions. The interactions of between bands V_1 with V_1 and V_2 are shown. Results for the four L-valleys in the Brillouin zone at the high-symmetry L-points are computed separately: (a-d) [1 1 1], (e-h) [$\bar{1}$ 1 1], (i-l) [1 $\bar{1}$ 1], (m-p) [1 1 $\bar{1}$].

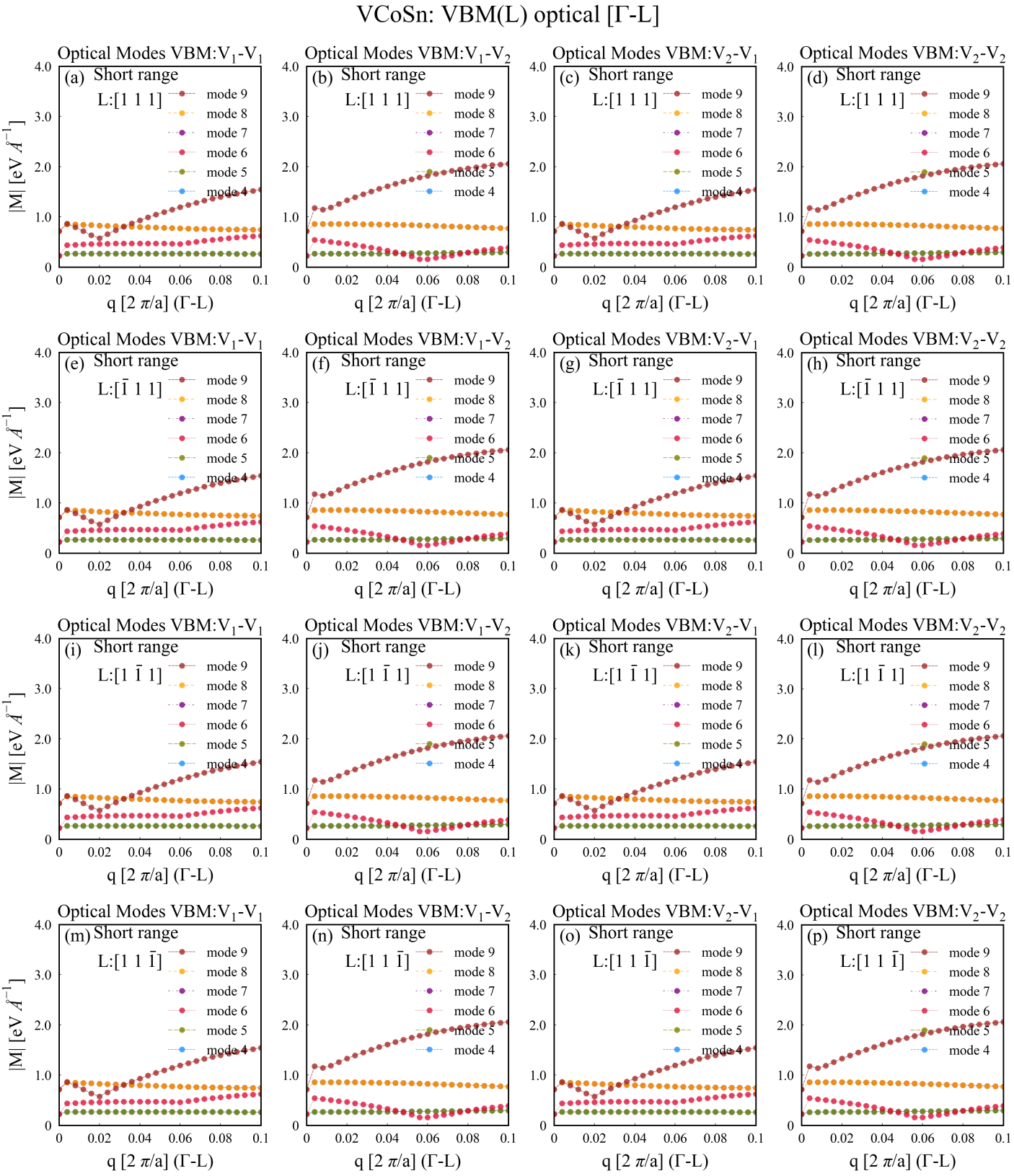


Figure 11: Short range matrix elements due to interactions of carriers with optical phonons for VCoSn around the VBM (at L) in the Γ-L direction, corresponding to intra- and inter-band transitions. The interactions of bands V_1 with V_1 and V_2 are shown. The four L-valley points in the Brillouin zone positionsare considered separately: (a-d) [1 1 1], (e-h) [$\bar{1}$ 1 1], (i-l) [1 $\bar{1}$ 1], (m-p) [1 1 $\bar{1}$].

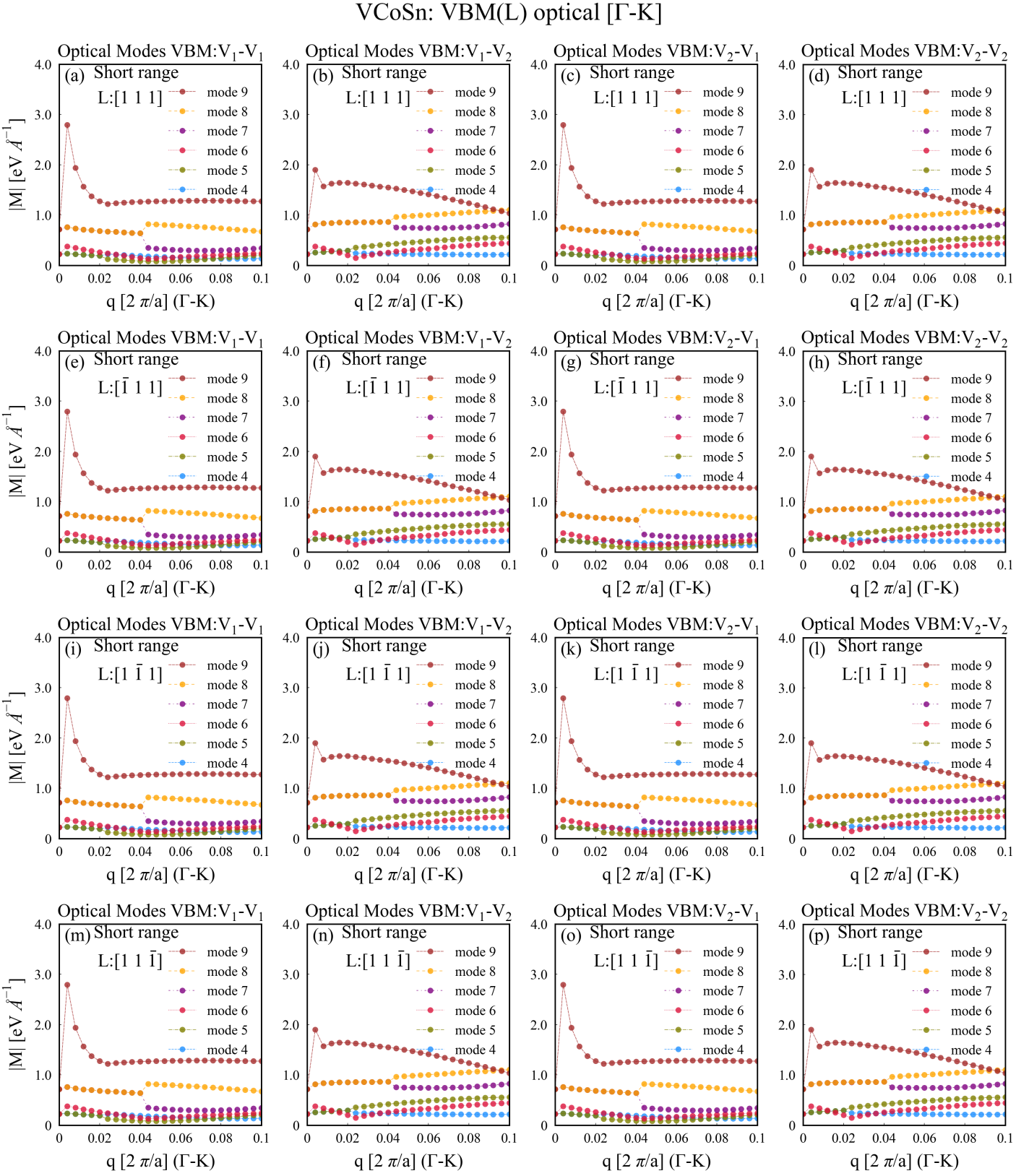


Figure 12: Short range matrix elements due to interactions of carriers with optical phonons for VCoSn around the VBM (at L) in the Γ-K direction, corresponding to intra- and inter-band transitions. The interactions of bands V_1 with V_1 and V_2 are shown. Four different equivalent L-point valleys are considered separately: (a-d) [1 1 1], (e-h) [$\bar{1}$ 1 1], (i-l) [1 $\bar{1}$ 1], (m-p) [1 1 $\bar{1}$].

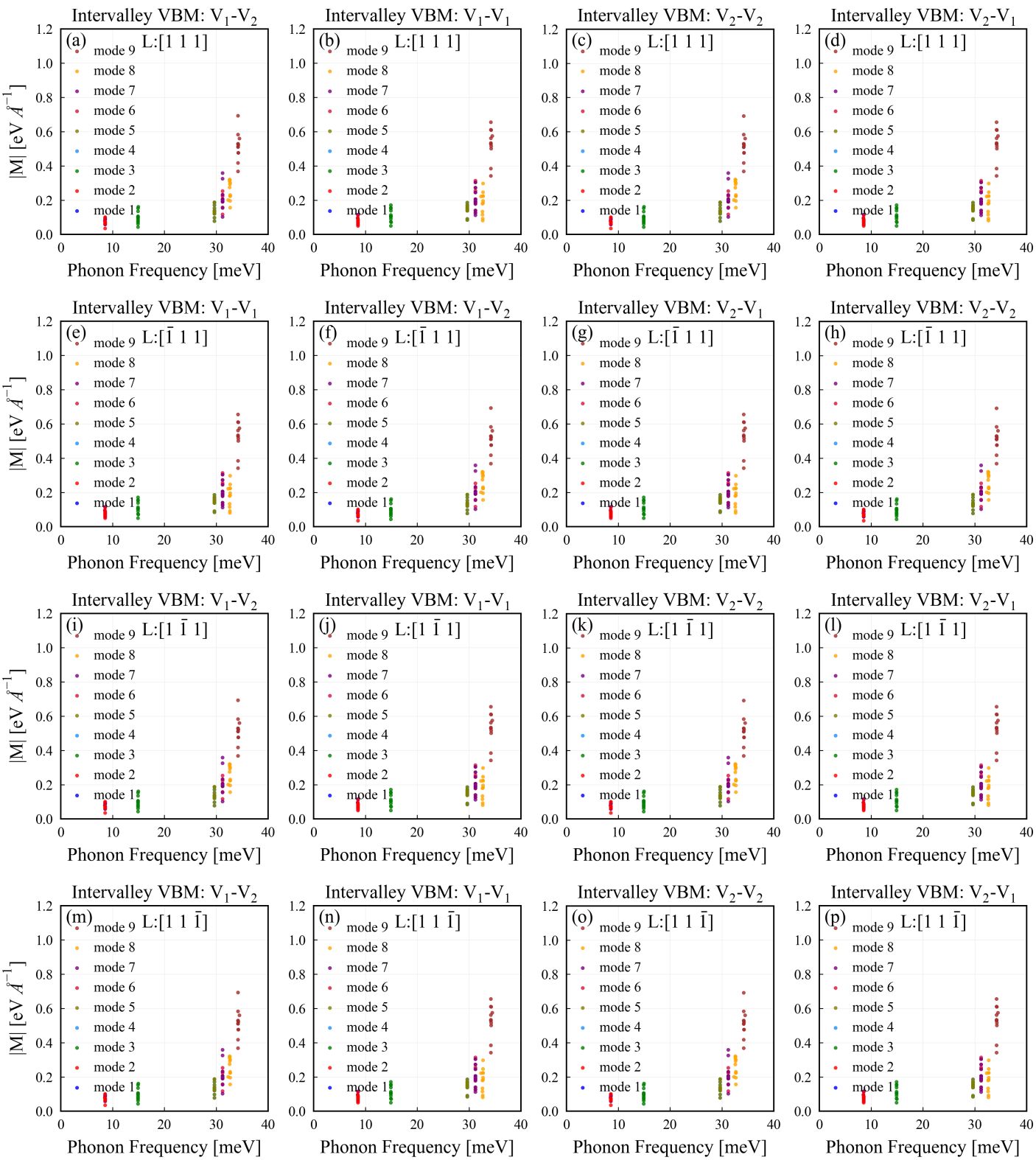


Figure 13: Short range matrix elements due to inter-valley interactions of carriers mediated with all 9 phonon modes for VCoSn at the VBM. Four different positions L-valley positions in the Brillouin zone are considered separately row-wise: (a-d) [1 1 1], (e-h) [$\bar{1}$ 1 1], (c) [0 0 1], (i-l) [1 $\bar{1}$ 1], ((m-p) [1 1 $\bar{1}$].

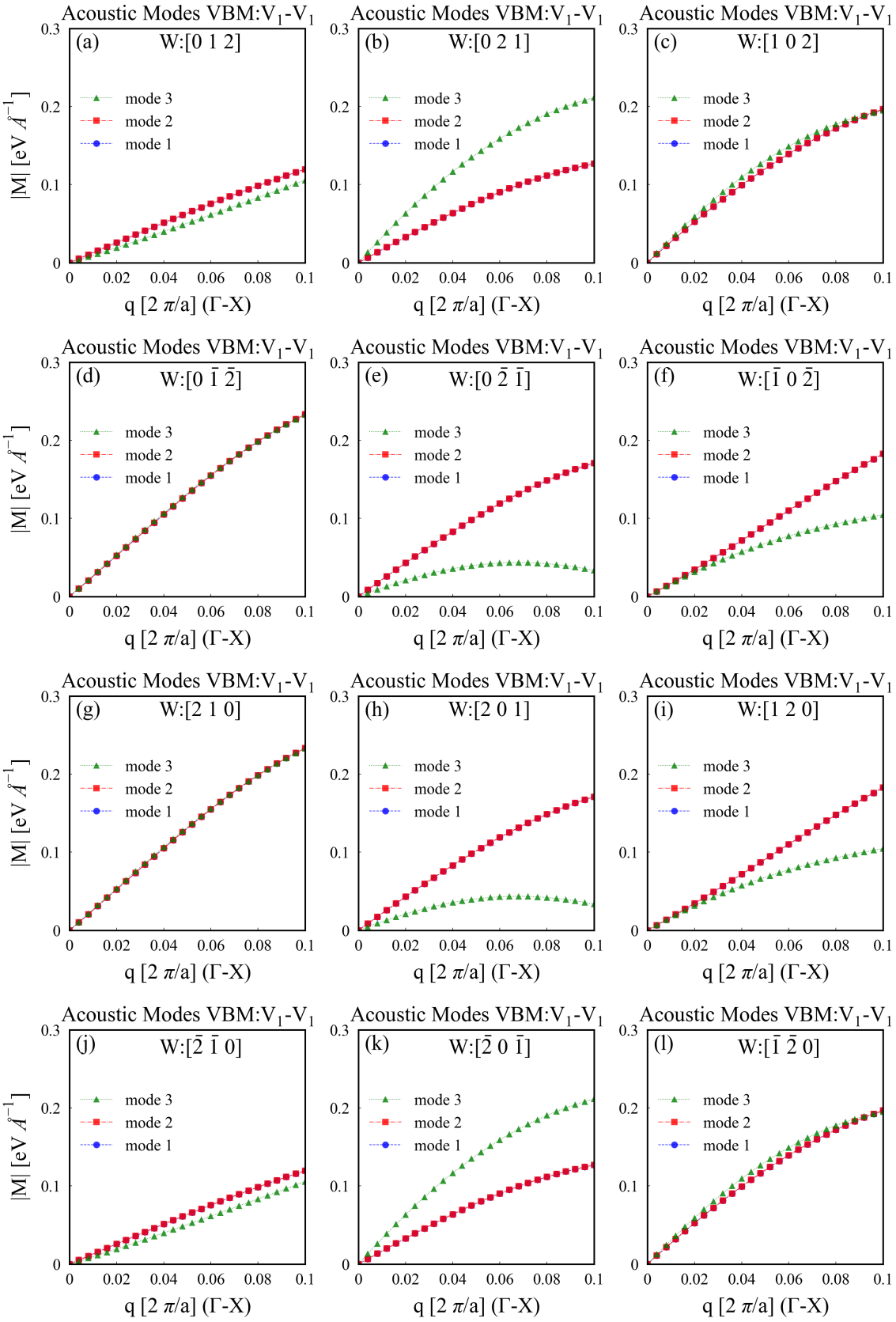


Figure 14: Matrix elements due to interactions of carriers around the VBM (at W) of CoVSn with acoustic phonons in the Γ-X direction, corresponding to intra-band transitions. The different sub-figures show results for interactions in the different identical positions of the high-symmetry W-points: (a) [0 1 2] (b) [0 2 1] (c) [1 0 2] (d) [0 $\bar{1}$ $\bar{2}$] (e) [0 $\bar{2}$ $\bar{1}$] (f) [$\bar{1}$ 0 $\bar{2}$] (g) [0 2 1] (h) [2 0 1] (i) [1 2 0] (j) [$\bar{2}$ $\bar{1}$ 0] (k) [$\bar{2}$ 0 $\bar{1}$] (l) [$\bar{1}$ $\bar{2}$ 0].

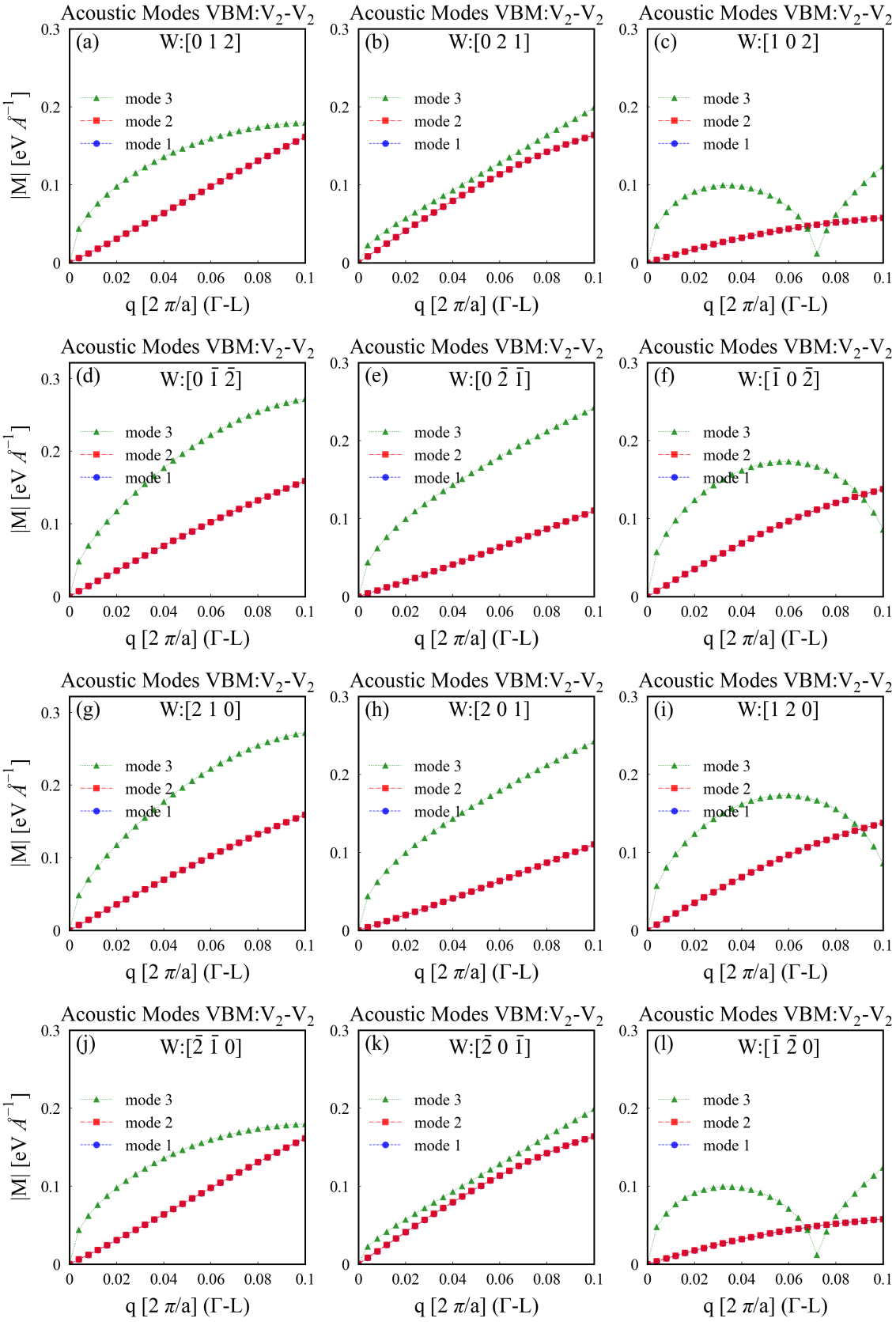


Figure 15: Matrix elements due to interactions of carriers around the VBM (at W) of CoVSn with acoustic phonons in the Γ-L direction, corresponding to intra-band transitions. The different sub-figures show results for interactions in the different identical positions of the high-symmetry W-points: (a) [0 1 2] (b) [0 2 1] (c) [1 0 2] (d) [0 $\bar{1}$ $\bar{2}$] (e) [0 $\bar{2}$ $\bar{1}$] (f) [$\bar{1}$ 0 $\bar{2}$] (g) [0 2 1] (h) [2 0 1] (i) [1 2 0] (j) [$\bar{2}$ $\bar{1}$ 0] (k) [$\bar{2}$ 0 $\bar{1}$] (l) [$\bar{1}$ $\bar{2}$ 0].

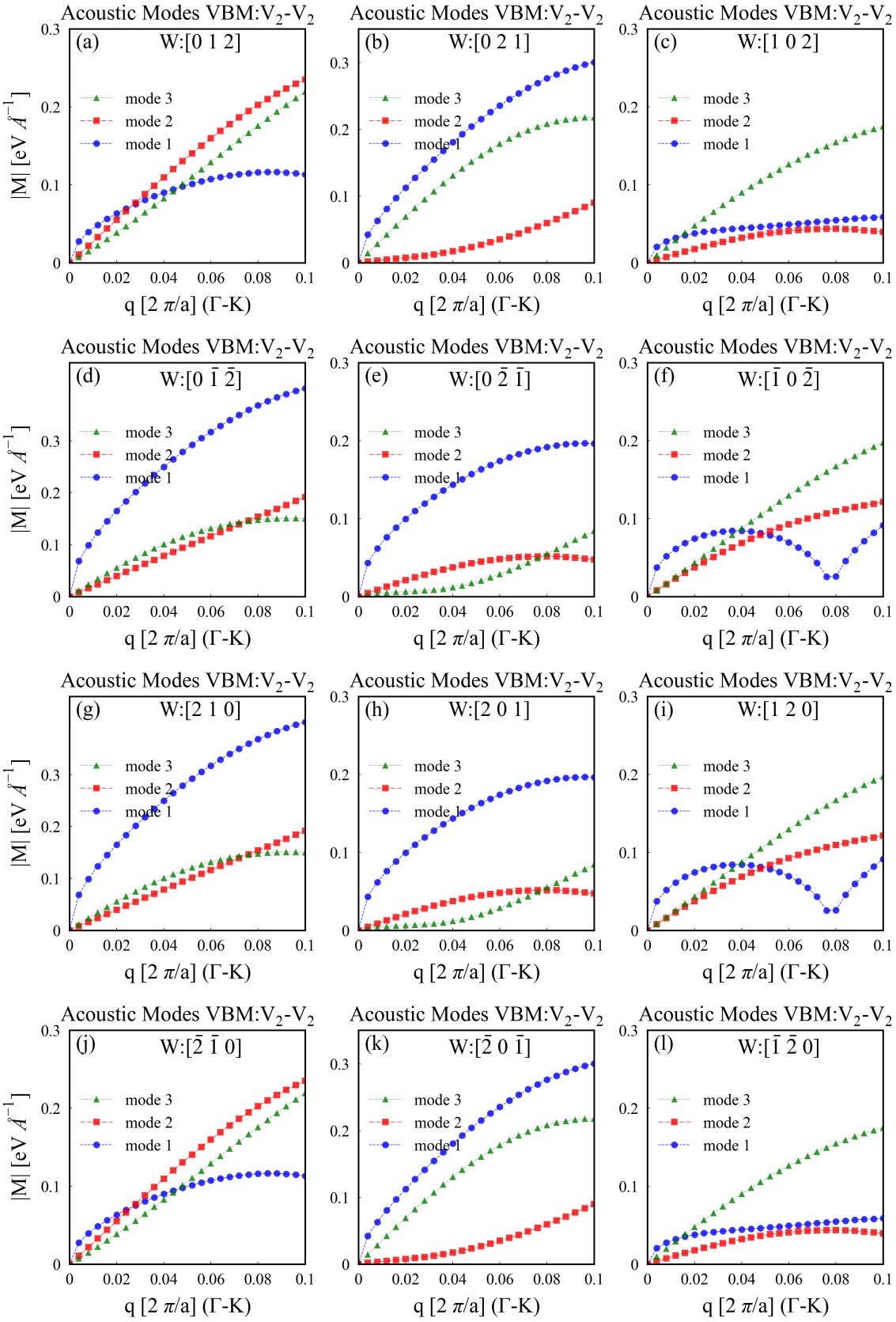


Figure 16: Matrix elements due to interactions of carriers around the VBM (at W) of CoVSn with acoustic phonons in the Γ-K direction, corresponding to intra-band transitions. The different sub-figures show results for interactions in the different identical positions of the high-symmetry W-points: (a) [0 1 2] (b) [0 2 1] (c) [1 0 2] (d) [0 $\bar{1}$ $\bar{2}$] (e) [0 $\bar{2}$ $\bar{1}$] (f) [$\bar{1}$ 0 $\bar{2}$] (g) [0 2 1] (h) [2 0 1] (i) [1 2 0] (j) [$\bar{2}$ $\bar{1}$ 0] (k) [$\bar{2}$ 0 $\bar{1}$] (l) [$\bar{1}$ $\bar{2}$ 0].

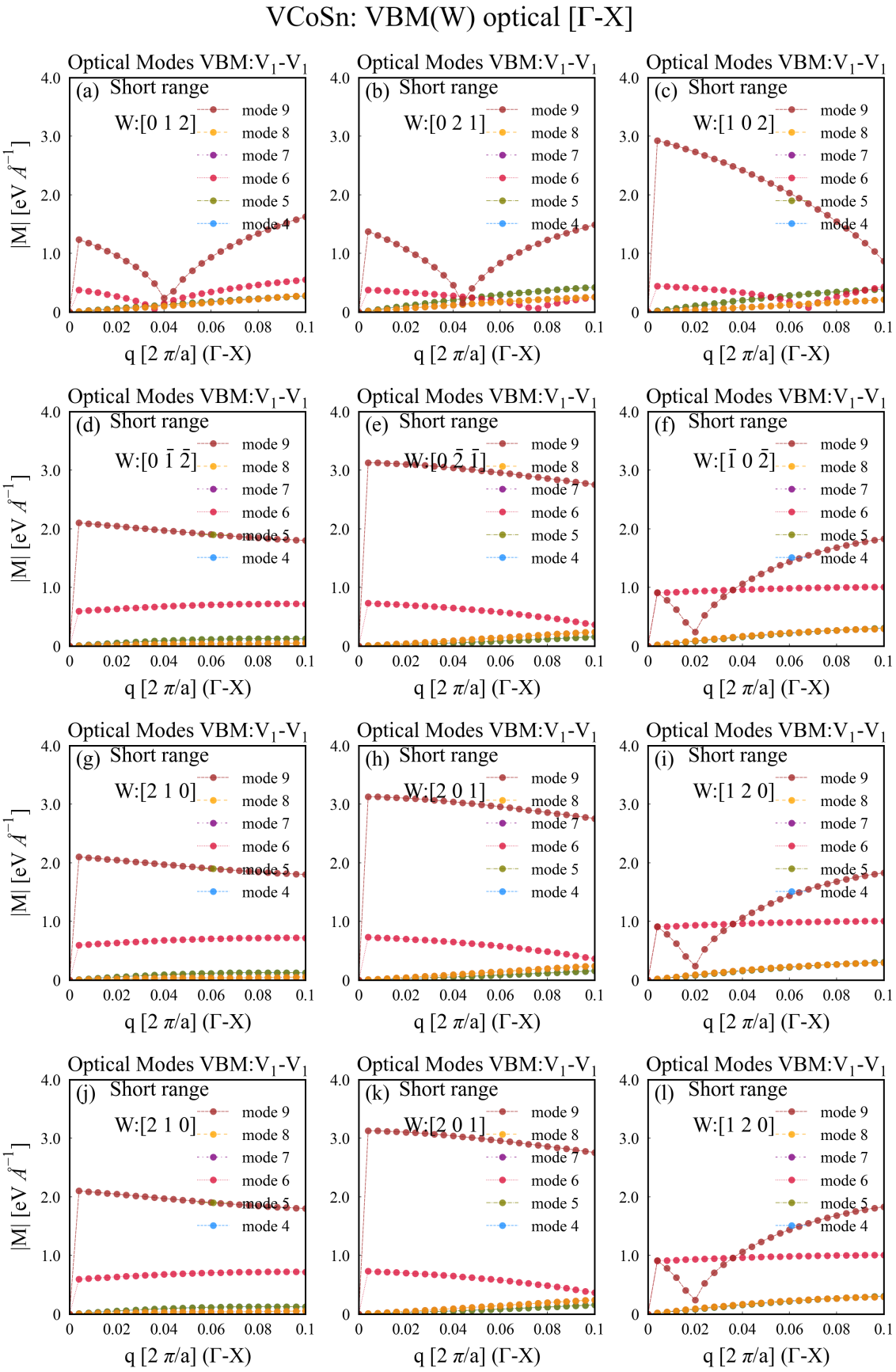


Figure 17: Matrix elements due to interactions of carriers around the VBM (at W) of CoVSn with optical phonons in the Γ-X direction, corresponding to intra-band transitions. The different sub-figures show results for interactions in the different identical positions of the high-symmetry W-points: (a) [0 1 2] (b) [0 2 1] (c) [1 0 2] (d) [0 $\bar{1}$ $\bar{2}$] (e) [0 $\bar{2}$ $\bar{1}$] (f) [$\bar{1}$ 0 $\bar{2}$] (g) [0 2 1] (h) [2 0 1] (i) [1 2 0] (j) [$\bar{2}$ $\bar{1}$ 0] (k) [$\bar{2}$ 0 $\bar{1}$] (l) [$\bar{1}$ $\bar{2}$ 0].

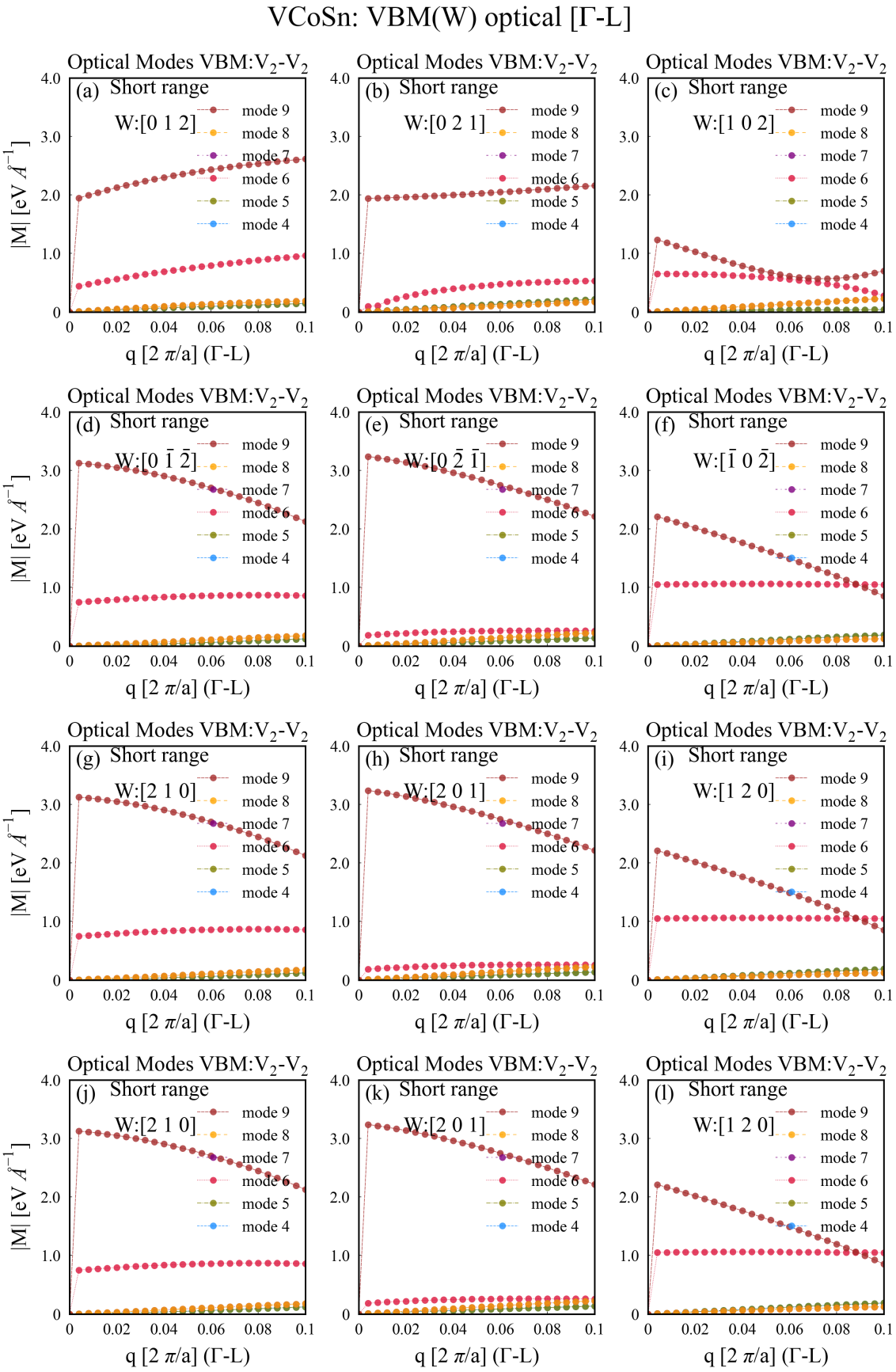


Figure 18: Matrix elements due to interactions of carriers around the VBM (at W) of CoVSn with optical phonons in the Γ-L direction, corresponding to intra-band transitions. The different sub-figures show results for interactions in the different identical positions of the high-symmetry W-points: (a) [0 1 2] (b) [0 2 1] (c) [1 0 2] (d) [0 $\bar{1}$ $\bar{2}$] (e) [0 $\bar{2}$ $\bar{1}$] (f) [$\bar{1}$ 0 $\bar{2}$] (g) [0 2 1] (h) [2 0 1] (i) [1 2 0] (j) [$\bar{2}$ $\bar{1}$ 0] (k) [$\bar{2}$ 0 $\bar{1}$] (l) [$\bar{1}$ $\bar{2}$ 0].

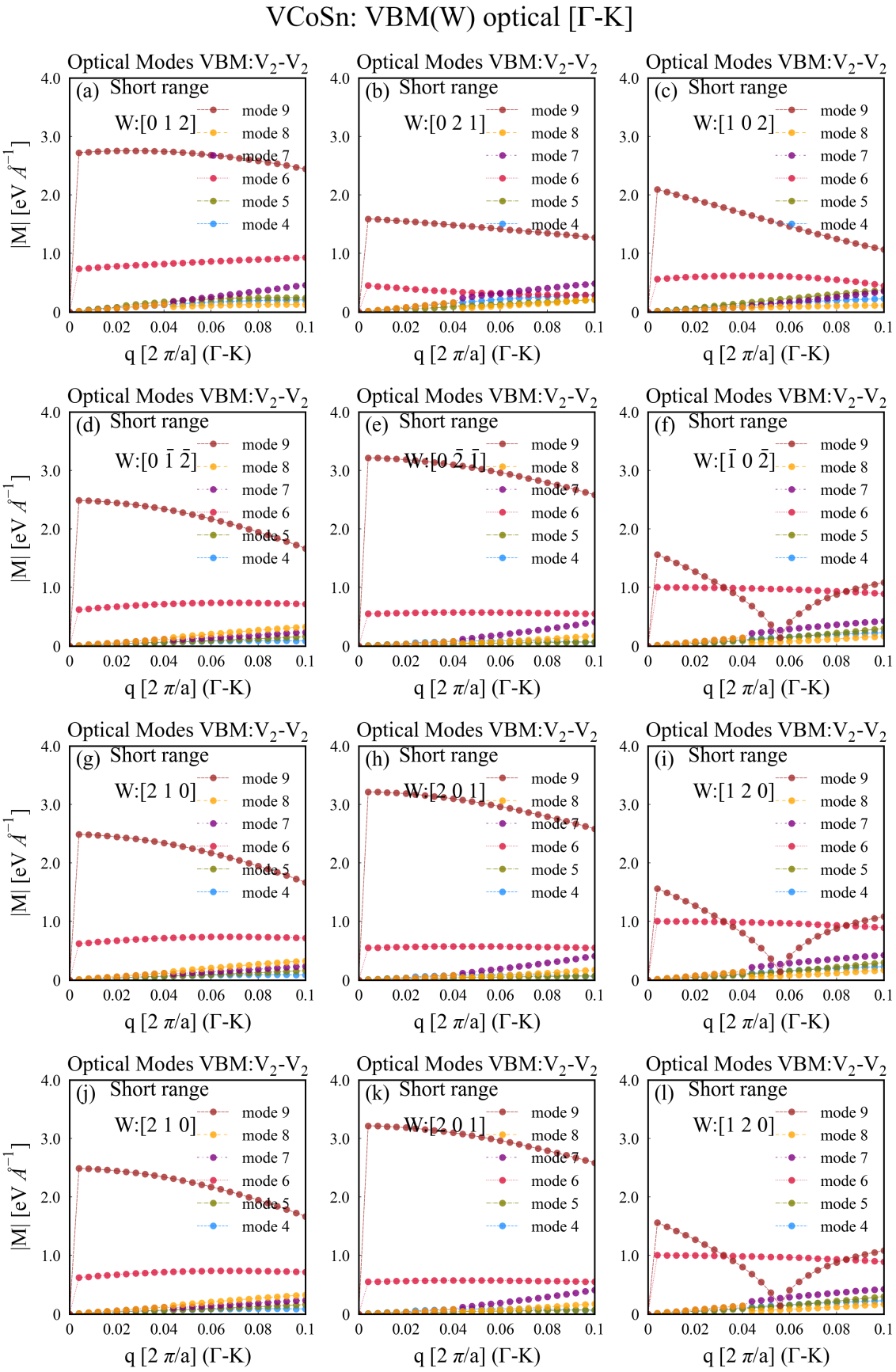


Figure 19: Matrix elements due to interactions of carriers around the VBM (at W) of CoVSn with optical phonons in the Γ-K direction, corresponding to intra-band transitions. The different sub-figures show results for interactions in the different identical positions of the high-symmetry W-points: (a) [0 1 2] (b) [0 2 1] (c) [1 0 2] (d) [0 $\bar{1}$ $\bar{2}$] (e) [0 $\bar{2}$ $\bar{1}$] (f) [$\bar{1}$ 0 $\bar{2}$] (g) [0 2 1] (h) [2 0 1] (i) [1 2 0] (j) [$\bar{2}$ $\bar{1}$ 0] (k) [$\bar{2}$ 0 $\bar{1}$] (l) [$\bar{1}$ $\bar{2}$ 0].